CONTRIBUTION JOURNAL: **Module 08: Bias–Variance Tradeoff**

ITAI 1371

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My Contributions - I used Google Colab, Python, NumPy, Matplotlib, and Scikit-learn to complete Lab 08. I generated a synthetic sine-wave dataset with random noise and trained polynomial regression models with degrees 1, 4, and 15. I plotted the three models to compare underfitting, a reasonable fit, and potential overfitting. I also created and used a plot_learning_curve() function to compare training and validation $R^2$ scores for degrees 1 and 15.

Challenges and Results - I corrected syntax and indentation errors and resolved a NameError by defining the function before calling it. I successfully generated the model-comparison graph and both learning curves. I observed extremely negative validation scores for the degree 15 model, suggesting poor generalization and possible numerical instability.

Conclusion - my contribution included coding, visualization, debugging, interpreting results, and answering the Knowledge Check questions. This lab strengthened my Python skills and understanding of model complexity, bias, variance, and generalization.

L08 – Contributions – Alexander DeBusk

**Module 08 – Contributions**

- Corrected and organized the Python code to ensure the imports, model training, and plotting functions worked properly.
- Created polynomial regression models with degrees 1, 4, and 15 to compare underfitting, a good fit, and overfitting.
- Implemented learning curves for the underfit and overfit models using five-fold cross-validation.
- Reviewed the plots and analyzed the differences between training scores and cross-validation scores.
- Completed the knowledge check by explaining model complexity, high bias, high variance, and the importance of generalization.
- Prepared the reflective journal discussing the challenges, lessons learned, and applications of the bias-variance tradeoff in machine learning and generative AI.
- Correctly saved all deliverables as pdf with appropriate naming conventions and submitted to group leader on time.

# Contribution

Through this lab, I learned several ways to improve machine learning models and help them make better predictions on new data. A more complex model does not always give better results. I should start with a simple model and adjust its complexity based on the results. Cross validation can help me compare different models and choose one that balances learning the training data by making accurate predictions on new data. Learning curves can help me identify underfitting and overfitting. If a model is underfitting, I can try a more complex model or add useful features. If a model is overfitting, I can reduce its complexity, use regularization, or collect more training data. Data quality is also important for model performance. Removing incorrect data, handling outliers, and selecting useful features can help a model learn meaningful patterns instead of noise. Preparing the data carefully can improve the model's performance. This lab helped me understand how polynomial regression and learning curves can be used to evaluate machine learning models. I learned that building a good model is not just about getting a high training score. It is also important to find a balance that allows the model to make reliable predictions on new data.